

Basic Circuits

Integrated circuits

Logic gates are never sold as individual, standalone components.

Instead, many gates are fabricated together on a single piece of silicon and packaged inside a protective plastic (or ceramic) casing

This is an **integrated circuit (IC)**, or "chip."

Integrated circuits

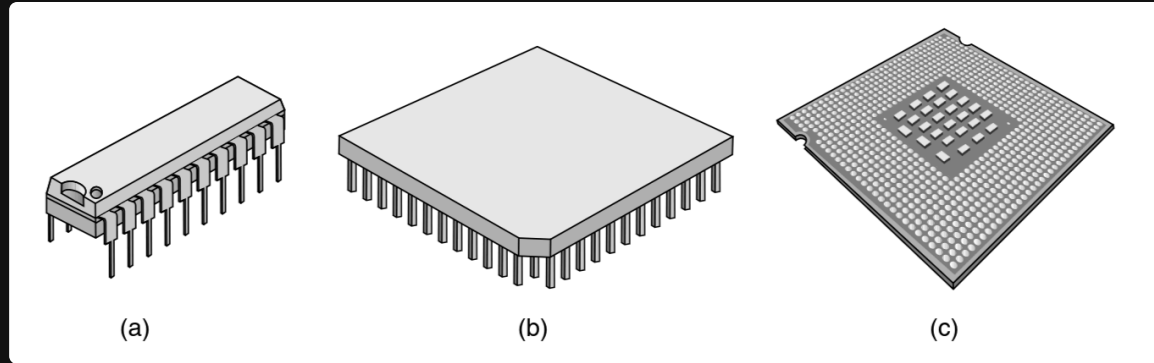
The silicon die itself is tiny

- the plastic package exists to protect it and to give it usable connection points.

Pins extending from the package are how the chip connects to the outside world

- some pins carry input signals,
- some carry output signals, and
- others supply power and ground.

Integrated circuits



Small/simple ICs (a handful of gates) typically use packages like **DIP (Dual In-line Package)**, with two rows of pins along the long edges.

Larger, more complex ICs (like CPUs) need far more connections than a two-row package can offer, so they use denser packaging styles:

- **PGA (Pin Grid Array)** - pins arranged in a grid on the underside of the package, plugging into a socket.
- **LGA (Land Grid Array)** - flat metal contact pads instead of pins, with the pins living in the socket instead.

Number of pins

Modern CPUs contain on the order of **1 billion transistors**.

Step 1 - Transistors -> Gates

Using a simplified estimate of 2 transistors per NAND gate:

$$1,000,000,000 \text{ transistors} \div 2 \approx 500,000,000 \text{ NAND gates}$$

Step 2 - Gates -> Pins

Each 2-input NAND gate needs 3 connections (2 inputs + 1 output). If every gate were its own individually packaged chip, we'd need:

$$500,000,000 \times 3 = 1,500,000,000 \text{ pins}$$

Add 2 more pins for shared power and ground:

$$1,500,000,002 \text{ pins total}$$

We **cannot** use raw inputs and outputs for pins